

Student Name : _____

Teacher Name : _____



Saint Ignatius' College, Riverview Mathematics Assessment Task 2025

Year 12
Extension 2 Mathematics
Task 4
Trial Higher School Certificate Examination
Date: 27 th August 2025, 1:00 pm

General Instructions:

- **Reading time:** 10 mins
- **Time Allowed:** 3 hours
- Write using blue or black pen only
- Board approved calculators may be used
- Attempt all questions in the space provided on the paper
- Write **your name** and your teachers initial on the cover of each booklet
- Marks may not be awarded for missing or carelessly arranged working
- A reference sheet is provided as a separate booklet.
- For questions in Section II, show relevant mathematical reasoning and/ or calculations

Teachers:

- | | |
|----------------|-----|
| • Mr R Maxwell | REM |
| • Mr D Reidy | DPR |

Section 1 10 Marks

Multiple Choice

- Attempt Questions 1-10
- Allow about 15 minutes for this section

Section 2 90 Marks

Extended Response

- Attempt Questions 11-16
- Allow about 2 hours and 45 minutes for this section

Total	100 Marks
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SECTION I

10 marks

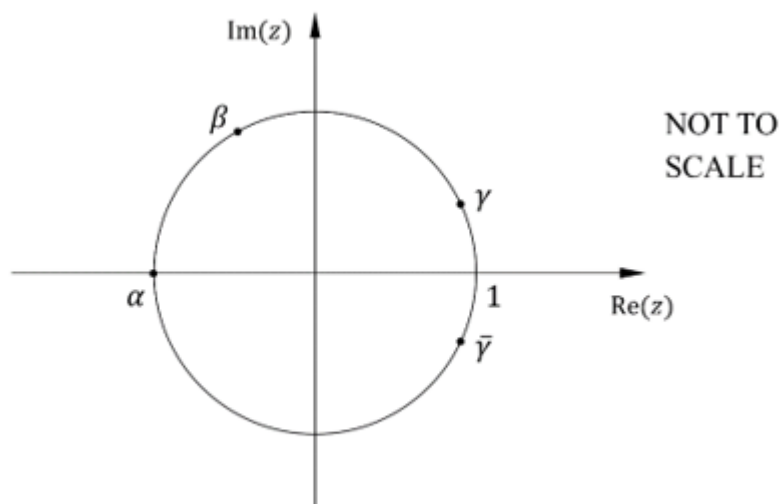
Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10

1. The points A, B, C are collinear where $\overrightarrow{OA} = \underline{i} + \underline{j}$, $\overrightarrow{OB} = 2\underline{i} - \underline{j} + \underline{k}$, $\overrightarrow{OC} = 3\underline{i} + a\underline{j} + b\underline{k}$.
What are the values of a and b ?
- A. $a = -3, b = -2$
- B. $a = 3, b = -2$
- C. $a = -3, b = 2$
- D. $a = 3, b = 2$
2. The negation of the statement $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}$ such that $2x + 3y = 12$ is :
- A. $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}$ such that $2x + 3y \neq 12$;
- B. $\exists x \in \mathbb{R}, \forall y \in \mathbb{R}$ $2x + 3y \neq 12$;
- C. $\forall x \in \mathbb{R}, \forall y \in \mathbb{R}$ $2x + 3y \neq 12$;
- D. $\exists x \in \mathbb{R}, \exists y \in \mathbb{R}$ such that $2x + 3y \neq 12$.
3. Which of the following is the converse of $\sim P \Rightarrow Q$?
- A. $\sim Q \Rightarrow P$
- B. $Q \Rightarrow \sim P$
- C. $P \Rightarrow Q$
- D. $\sim P \Rightarrow \sim Q$

4. Four roots of the polynomial $P(x)$ lie on the unit circle on a complex plane, as shown below. There may be other roots.

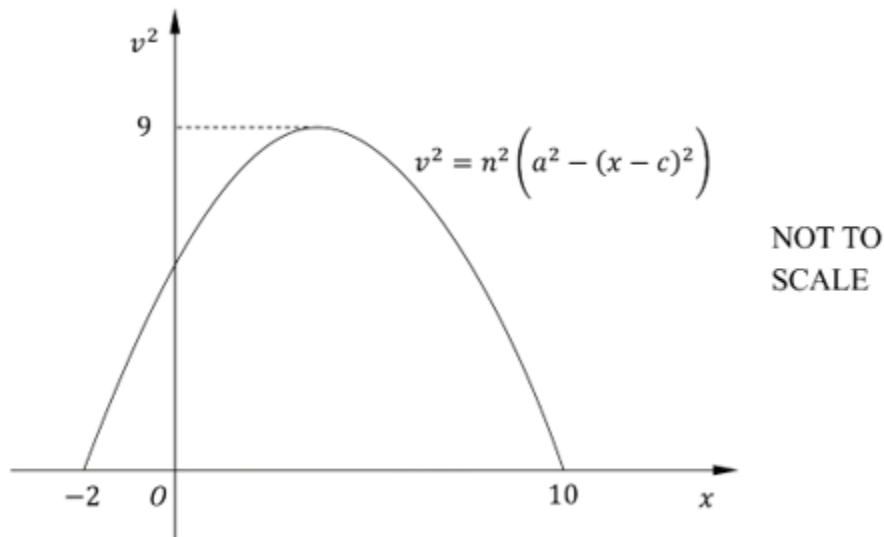


Which statement about $P(x)$ MUST be True.

- A. If $P(x)$ is a quartic then all its coefficients are real.
- B. If $P(x)$ is a quartic then at least one of its coefficients is real.
- C. If all of the coefficients of $P(x)$ are real then its degree is at least 5.
- D. $P(x)$ is in the form $z^n + 1 = 0$ where n is odd and at least 5.

5. A particle is moving along the x -axis in simple harmonic motion. The displacement of the particle is x metres and its velocity is $v \text{ ms}^{-1}$

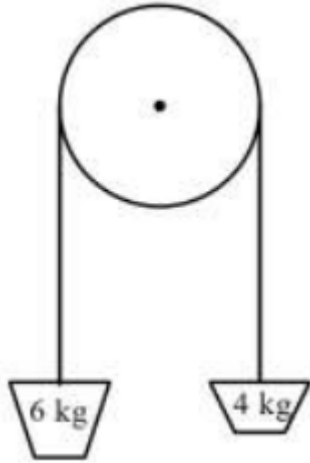
The graph of $v^2 = n^2 (a^2 - (x - c)^2)$, where a , c and n are positive constants, is shown below.



The values of a , c and n are :-

- A. $a = 4$, $c = 6$ and $n = \frac{1}{2}$
- B. $a = 4$, $c = 6$ and $n = \frac{3}{2}$
- C. $a = 6$, $c = 4$ and $n = \frac{1}{2}$
- D. $a = 6$, $c = 4$ and $n = \frac{3}{2}$

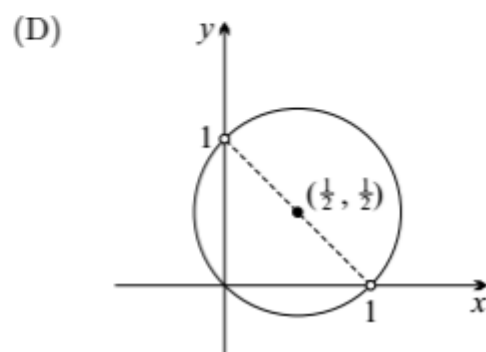
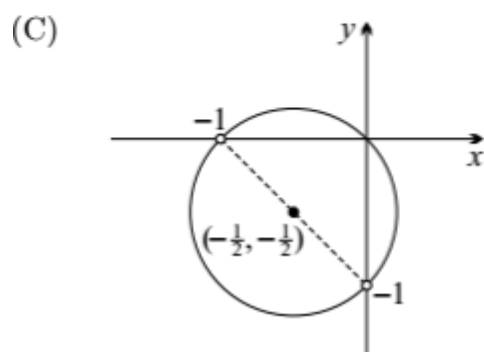
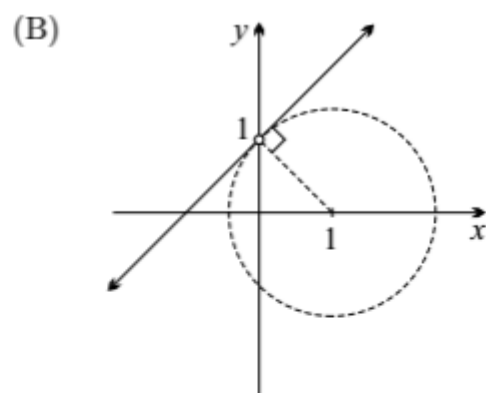
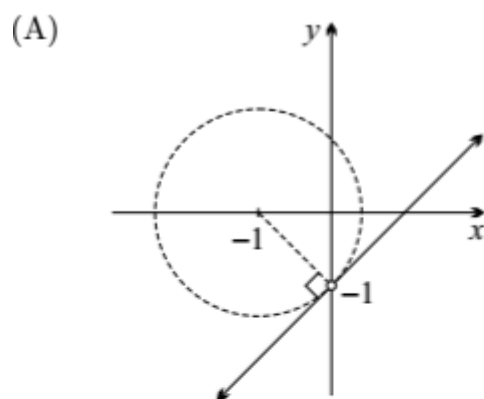
6. Particles of mass 6 kg and 4 kg are attached to each end of a light inextensible string. The string passes over a pulley as shown in the diagram below.



Which expression gives the acceleration of the 6 kg mass as it moves downwards ?

- A. $\frac{1}{5}g$
- B. $\frac{2}{5}g$
- C. $2.5g$
- D. $5g$

7. If $\omega = \frac{z+1}{z+i}$ and ω is purely imaginary, what is the locus of z ?



8. Which of the following is equivalent to $\frac{e^{-\frac{i\pi}{2}}}{e^{\frac{i\pi}{6}}}$?

A. $-\frac{1}{2} - \frac{\sqrt{3}}{2}i$

B. $-\frac{1}{2} + \frac{\sqrt{3}}{2}i$

C. $\frac{1}{2} - \frac{\sqrt{3}}{2}i$

D. $\frac{1}{2} + \frac{\sqrt{3}}{2}i$

9. The complex number $z = a + ib$, where $0 < a < b$.

Which of the following best describes the complex number z^4 ?

- A. $\operatorname{Re}(z^4) < 0$
- B. $\operatorname{Re}(z^4) \leq 0$
- C. $\operatorname{Im}(z^4) < 0$
- D. $\operatorname{Im}(z^4) \leq 0$

10. Suppose there exist vectors $\vec{a}$, $\vec{b}$ and $\vec{r}$ such that $\vec{r} = \vec{a} + (\vec{a} \cdot \vec{b}) \vec{b}$.

If $\vec{r}$ is perpendicular to $(\vec{a} - \vec{b})$, the minimum value of $1 + |\vec{b}|^2$ is ?

- A. $|\vec{a}| + 1$
- B. $\frac{|\vec{a}| + 1}{2}$
- C. $2|\vec{a}|$
- D. $|\vec{a}|$

END OF SECTION 1

SECTION II**90 marks****Attempt Questions 11–16****Allow about 2 hours and 45 minutes for this section**Answer each question in a **SEPARATE** writing booklet. Extra writing booklets are available.

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations

	Marks
Question 11 (15 marks) Use a SEPARATE writing booklet.	
(a) Find $\int x e^{-2x} dx$.	2
(b) If $z = 2 + i$ and $w = 3 - 2i$, express in the form $x + iy$	
(i) $2z + \bar{w}$	1
(ii) $\frac{1}{z - w}$	1
(c) Consider the lines	
$l_1 : x = y = z$ $l_2 = \begin{pmatrix} 0 \\ 3 \\ 2 \end{pmatrix} + t \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$ where t is a parameter.	2
Find the angle between the two lines.	
(d) Evaluate $\int_0^{\frac{\pi}{4}} \frac{d\theta}{4 + 2\sin 2\theta}$, using the substitution, $t = \tan \theta$.	3

Question 11 continued on the next page

	Marks
(e) Consider the complex number $z = -1 + \sqrt{3} i$.	
(i) Write z in modulus-argument form.	2
(ii) Find the exact value of z^{10} in exact cartesian form.	1
(f) The locus of a point P which moves in the complex plane is represented by the equation $ z - 3 - 4i = 5$.	
(i) Sketch the locus of the point P , noting intercepts.	2
(ii) Find the exact value of $\arg z$ when P is in the position of maximum modulus.	1

End of Question 11

Question 12 (14 marks) Use a SEPARATE writing booklet.

- (a) Find the point(s) of intersection of the line with parametric equation

$$\vec{r} = \vec{i} + 3\vec{j} - 4\vec{k} + t \left(\vec{i} + 2\vec{j} + 2\vec{k} \right), \text{ with the sphere with equation}$$

$$(x-1)^2 + (y-3)^2 + (z+4)^2 = 81$$

3

- (b) Use partial fraction to find

$$\int \frac{x^2 + 8x + 11}{(x-3)(x^2+2)} dx$$

3

- (c) For s , a real variable, the complex number z moves on the Argand plane according to the formulae $z = \frac{1}{1+is}$ as s varies from $-\infty$ to $+\infty$.

- (i) Find the value of s for which $z = \frac{1}{2} + \frac{1}{2}i$.

1

- (ii) The locus of z is, in fact, a circle. Find the cartesian equation of the locus, writing down its centre and radius.

3

- (d) The equation of two lines are

$$\vec{r}_1 = \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 4 \\ 1 \end{pmatrix} \quad \text{and} \quad \vec{r}_2 = \begin{pmatrix} -4 \\ 6 \\ -2 \end{pmatrix} + \mu \begin{pmatrix} 4 \\ -11 \\ 3 \end{pmatrix}$$

These lines are not parallel so determine if they skew or whether they intersect, showing all necessary working.

2

Question 12 continued on the next page

Marks

- (e) Show that for $x \in \mathbb{Z}$, if $x^2 - 6x + 5$ is even, then x is odd.

2

End of Question 12

Question 13 (16 marks) Use a SEPARATE writing booklet**Marks**

- (a) Find the point on the line with vector equation $\vec{r} = \begin{pmatrix} 1 \\ 4 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}$ that is closest to the origin. **3**

- (b) A particle is moving along the x axis in simple harmonic motion centred at $x = 2$ and with amplitude 5.

The velocity of the particle is given by $v^2 = p + 8x - qx^2$, where p and q constants.

Find the exact numerical value for the period of the motion.

3

- (c) A particle is moving in the positive direction along a straight line in a medium that exerts a resistance to motion proportional to the cube of the velocity.

No other forces act on the particle, that is $\ddot{x} = -kv^3$, where k is a positive constant.

At time $t = 0$, the particle is at the origin and has velocity U . At time $t = T$, the particle is at $x = D$ and has a velocity V .

- (i) Show that $\frac{1}{V^2} - \frac{1}{U^2} = 2kT$. **3**

- (ii) Also, show that $\frac{1}{V} - \frac{1}{U} = kD$. **3**

- (iii) Hence show that $\frac{D}{T} = \frac{2UV}{U+V}$. **1**

- (d) (i) Prove that $a^2 + b^2 + c^2 \geq ab + ac + bc$. **2**

- (ii) Hence, or otherwise prove that $a^2b^2 + a^2c^2 + b^2c^2 \geq abc(a + b + c)$ **1**

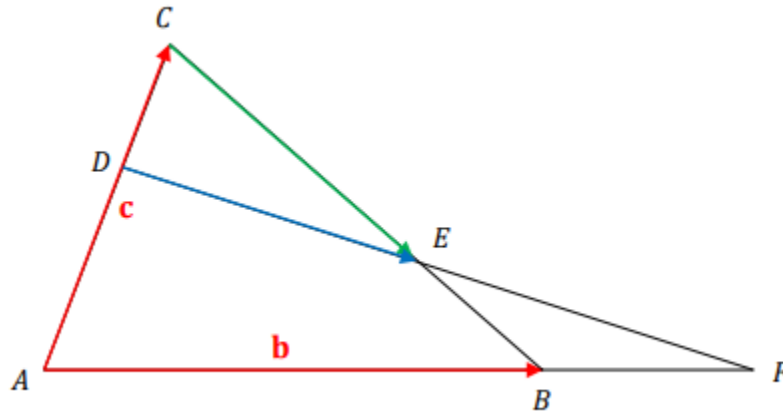
End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet

- (a) Suppose that n and $n + 1$ are positive integers, neither of which is divisible by 3. 2
Prove that $n^3 + (n + 1)^3$ is divisible by 9.
- (b) The complex number ω is given as $\omega = \frac{z - 1}{z - 2i}$. 2
Describe the locus of z geometrically if ω is real.
- (c) Suppose that p is a prime number and a is a positive integer.
- (i) Use the expansion of $(1 + a)^p$ to prove that $(1 + a)^p - 1 - a^p$ is divisible by p . 1
(You may use the fact that $\binom{p}{k}$ is divisible by p for $k = 1, 2, 3, \dots, p - 1$.)
- (ii) Hence prove by mathematical induction that $a^p - a$ is divisible by p for all positive integers values of a . 3
- (d) Find $\int \frac{x^2}{x + 1} dx$. 2

Question 14 continued on the next page

(e)



In $\triangle ABC$ above, D is the point on AC such that $AD : DC = 2 : 1$ and E is the point on BC such that $BE : EC = 1 : 2$. When DE is extended, it meets AB extended at F .

Let $\vec{AB} = b$ and $\vec{AC} = c$.

(i) Express $\vec{CE}$ in terms of b and c .

1

(ii) Show that $\vec{DE} = \frac{2}{3}b - \frac{1}{3}c$.

1

(iii) Show that $AF : FB = 4 : 1$

3

End of Question 14

Question 15 (15 marks) Use a SEPARATE writing booklet

Marks

(a) Let $I_n = \int_0^1 (1-x^2)^{\frac{n-1}{2}} dx$. where $n = 1, 2, 3, \dots$ show that

Show that $n I_n = (n-1) I_{n-2}$ for $n = 2, 3, 4, \dots$

3

(b) A particle of mass m is moving in a straight line under the action of a force F where $F = \frac{m}{x^3} (6 - 10x)$ and x is its displacement from the origin.

(i) Find its velocity if the particle starts from rest at $x = 1$.

3

(ii) At what other position does the particle come to rest.

1

(c) Find $\int_1^2 \frac{x+1}{\sqrt{-2+3x-x^2}} dx$

3

Question 15 continued on the next page

(d) Consider the line $r = \lambda \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$.

(i) Show that the position vector of the point on the line $r_{\sim}$ closest to the point (x_0, y_0, z_0) is :

$$\frac{x_0 + 2y_0 + z_0}{6} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}.$$

2

(ii) Hence, or otherwise, find the point on the line $r_{\sim}$ that is closest to a second line

3

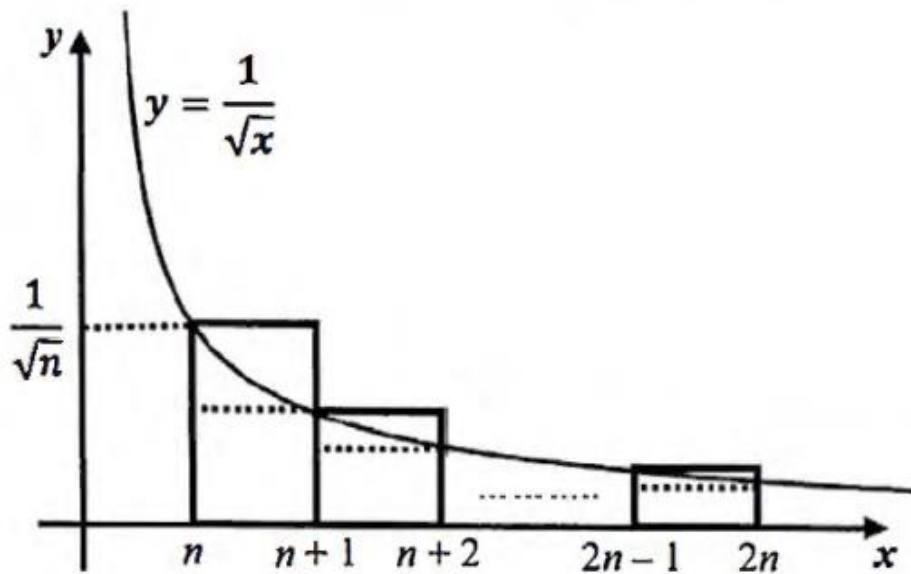
$$s_{\sim} = \begin{bmatrix} 4 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 1 \\ 5 \end{bmatrix}, \text{ where } t \in (-\infty, \infty).$$

End of Question 15

Question 16 (15 marks) Use a SEPARATE writing booklet.

Marks

- (a) In the diagram below, the graph of $y = \frac{1}{\sqrt{x}}$ has been drawn, and n rectangles have been constructed between $x = n$ and $x = 2n$, each of width 1 unit.



- (i) Show that $\int_n^{2n} \frac{dx}{\sqrt{x}} = 2\sqrt{n}(\sqrt{2} - 1)$.

1

- (ii) Let $S_n = \frac{1}{\sqrt{n+1}} + \frac{1}{\sqrt{n+2}} + \dots + \frac{1}{\sqrt{2n}}$

Show that ;

$$2\sqrt{n}(\sqrt{2} - 1) + \frac{1 - \sqrt{2}}{\sqrt{2n}} < S_n < 2\sqrt{n}(\sqrt{2} - 1)$$

4

Question 16 continued on the next page

- (b) A particle's movement is represented by the vector $\vec{r} = \begin{pmatrix} t + t^{-1} \\ t^3 + t^{-3} \end{pmatrix}$, where $t \in \mathbb{R}^+$

Sketch the Cartesian graph that shows the path the particle can move along.

3

- (c) A particle A of mass m falls vertically from rest at a height h above the ground. It is Subject to a resistance mkv where v is its speed and k is a positive constant. The acceleration due to gravity is g .

- (i) Show that $\ddot{x} = k(U - v)$, where U is the terminal speed of the particle.

1

- (ii) Prove that the speed of particle A at any time t is given by $v = U(1 - e^{-kt})$

3

- (iii) At the same time as particle A falls from rest another particle B of mass m is projected upwards in the same vertical line as particle A with a speed of kU , where U is the terminal speed of the particle. Hence it can be shown that for particle B its speed at any time t is given by

$$v = kU e^{-kt} - U(1 - e^{-kt}) \quad . \quad (\text{DO NOT PROVE THIS})$$

If the two particles collide after time T seconds when particle A has fallen a distance of X , prove that

3

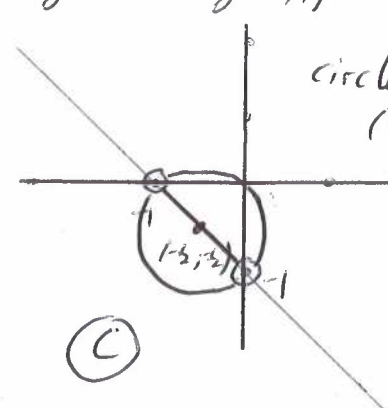
$$T = \frac{1}{k} \log_e \left(\frac{U}{U - h} \right)$$

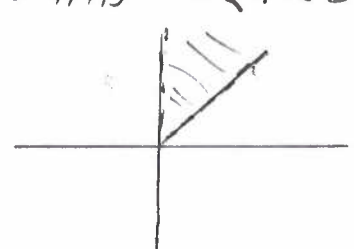
End of Exam

Year 12 - 2025 Term 3 Mathematics (Ext 2) Assessment Task 4 Trial HSC Exam Solutions

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
1. $\vec{OA} = \begin{bmatrix} 1 \\ 1 \\ 6 \end{bmatrix}$ $\vec{OB} = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$ $\vec{OC} = \begin{bmatrix} 3 \\ 9 \\ b \end{bmatrix}$ $\vec{AB} = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \\ 6 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \\ -5 \end{bmatrix}$ $\vec{BC} = \begin{bmatrix} 3 \\ 9 \\ b \end{bmatrix} - \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 10 \\ b-1 \end{bmatrix}$ $\begin{bmatrix} 1 \\ -2 \\ -5 \end{bmatrix} = \lambda \begin{bmatrix} 1 \\ 10 \\ b-1 \end{bmatrix}$ $1 = \lambda$ $-2 = 9\lambda + 1 \Rightarrow a = -3$ $1 = b\lambda - 1 \Rightarrow b = 2$ (C)		3. $\sim p \Rightarrow q$ Converse is $q \Rightarrow \sim p$ (B)	
2. $\forall x \in \mathbb{R}, \exists y \in \mathbb{R} \therefore$ $2x + 3y = 12$ Negation $\exists x \in \mathbb{R}, \forall y \in \mathbb{R}$ $2x + 3y \neq 12$ (B)		4. Roots are $y, \bar{y}$, $\alpha(\text{real}) = -1, \beta$ If coefficients of $P(x)$ are real then it degree must be at least 5 as $\bar{\beta}$ is needed (C) NOTE: p could be true but not <u>must</u> be true	
		5. $v^2 = n^2 (a^2 - (x-c)^2)$ $v=0 \quad x = -2 \text{ or } 10$ centre is $x = 4$ amplitude $a = 6$ max velocity at $x = c$ ie $v^2 = n^2 a^2$ $n^2 = \frac{9}{36} = \frac{1}{4} \therefore n = \frac{1}{2}$ $a = 6, c = 4, n = \frac{1}{2}$ (C)	

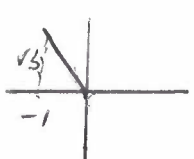
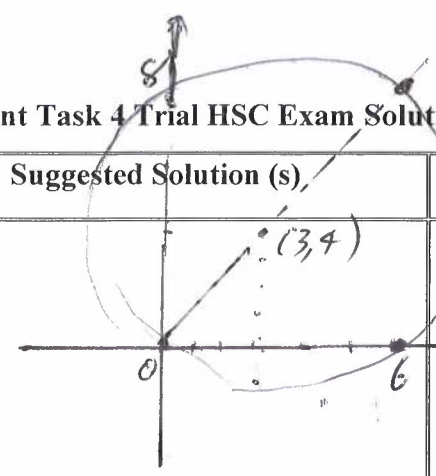
Year 12 - 2025 Term 3 Mathematics (Ext 2) Assessment Task 4 Trial HSC Exam Solutions

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
6. At $m = 6 \text{ kg}$ $F_{\text{net down}} = Mg - T$ $= 6g - T$ $\ddot{x} = 6g - T / 6$ At $m = 4 \text{ kg}$ $F_{\text{net up}} = T - Mg$ $= T - 4g$ $\ddot{x} = \frac{T - 4g}{4}$ $\frac{6g - T}{6} = \frac{T - 4g}{4}$ $24g - 4T = 6T - 24g$ $10T = 48g$ $T = \frac{48g}{10}$ $\ddot{x}_{\text{avg}} = \frac{6g - \frac{48g}{10}}{6}$ $= g - \frac{4}{5}g$ $= \frac{1}{5}g$ (A)		7. $w = \frac{z+1}{z+i}$ is purely imaginary $\arg(w) = \arg(z+1) - \arg(z+i) = \pm \frac{\pi}{2}$  circle centre $(-\frac{1}{2}, -\frac{1}{2})$ but not including $(-1, 0)$ or $(0, -1)$ (C)	
		8. $\frac{e^{-i\frac{\pi}{2}}}{e^{i\frac{\pi}{6}}}$ $= e^{-i\frac{\pi}{2} - i\frac{\pi}{6}} = e^{-4i\frac{\pi}{6}}$ $= e^{-2i\frac{\pi}{3}}$ $= \cos(-\frac{2\pi}{3}) + i\sin(-\frac{2\pi}{3})$ $= -\frac{1}{2} - \frac{\sqrt{3}}{2}i$ (A)	

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
9. $z = a + ib$ $0 \leq a < b$  $\frac{\pi}{4} < \arg(z) < \frac{\pi}{2}$ $\therefore \pi < \arg(z^4) < 2\pi$ $\therefore \operatorname{Im}(z^4) < 0$ (C)		$\therefore$ real solutions $\Delta = (1+ b ^2)^2 - 4 z ^2 > 0$ $1+ b ^2 > 2 z$ min value $2 z$ (C)	
10. $x \cdot b (x-b)$ $\therefore x \cdot (x-b) = 0$ $x \cdot x = x \cdot b$ Subbing $x = a + (a-b)b$ $a \cdot [a + (a-b)b] = b [a + (a-b)b]$ $a ^2 + (a-b)^2 = a \cdot b + (a-b) b ^2$ $= a \cdot b (1+ b ^2)$ $\therefore (a-b)^2 - (a-b)(1+ b ^2) + a ^2 = 0$ quadratic in $a-b$		11. (a) $\int x e^{-2x} dx$ $= x \cdot -\frac{1}{2} e^{-2x} - \int -\frac{1}{2} e^{-2x} dx$ ✓ $= -\frac{1}{2} x e^{-2x} + \frac{1}{2} \int e^{-2x} dx$ $= -\frac{1}{2} x e^{-2x} - \frac{1}{4} e^{-2x} + C$ OR $= -\frac{1}{4} e^{-2x} (2x+1) + C$ ✓	
		(b) $z = 2+i$ $w = 3-2i$ (i) $2z + \bar{w}$ $= 2(2+i) + (3+2i)$ $= 7+4i$ ✓	

Year 12 - 2025 Term 3 Mathematics (Ext 2) Assessment Task 4 Trial HSC Exam Solutions

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
(ii) $\frac{1}{z-w}$ $= \frac{1}{2+i - (3-2i)}$ $= \frac{1}{-1+3i} \times \frac{-1-3i}{-1-3i}$ $= \frac{-1-3i}{(-1)^2 - (3i)^2} = \frac{-1-3i}{1+9}$ $= -\frac{1}{10} - \frac{3}{10}i \quad \checkmark$		(d) $\int_0^{\frac{\pi}{4}} \frac{db}{4+2\sin 2\theta}$ $t = \tan \theta$ $dt = \sec^2 \theta d\theta$ $= (1+t^2) d\theta$ $d\theta = \frac{dt}{1+t^2}$ $\theta = \frac{\pi}{4} \quad t = 1$ $\theta = 0 \quad t = 0$ $I = \int_0^1 \frac{dt}{4+2\left(\frac{2t}{1+t^2}\right)} \frac{dt}{1+t^2} \quad \checkmark$ $= \int_0^1 \frac{dt}{4t^2 + 4t + 4}$ $= \frac{1}{4} \int_0^1 \frac{dt}{t^2 + t + 1}$ $= \frac{1}{4} \int_0^1 \frac{dt}{(t+\frac{1}{2})^2 + (\frac{\sqrt{3}}{2})^2}$ $= \frac{1}{4} \left[\frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{2t+1}{\sqrt{3}} \right) \right]_0^1$ $= \frac{1}{2\sqrt{3}} \left(\tan^{-1} \sqrt{3} - \tan^{-1} \left(\frac{1}{\sqrt{3}} \right) \right)$	
(c) $l_1: x=y=z$ $l_2: \begin{pmatrix} 0 \\ 3 \\ 2 \end{pmatrix} + t \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$ $\cos \alpha = \frac{\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}}{\sqrt{3} \sqrt{6}} = 0 \quad \checkmark$ $\therefore$ lines perpendicular $\therefore \alpha = 90^\circ \quad \checkmark$		$= \frac{1}{2\sqrt{3}} \left(\frac{\pi}{3} - \frac{\pi}{6} \right) = \frac{\pi}{12\sqrt{3}} \text{ or } \frac{\sqrt{3}\pi}{36}$	

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
e) $z = -1 + \sqrt{3}i$ (i) $z = \sqrt{(-1)^2 + (\sqrt{3})^2} = 2$ ✓ $\arg(z)$  $= \pi - \tan^{-1} \sqrt{3}$ $= \frac{2\pi}{3}$ ✓ $z = 2 \operatorname{cis} \frac{2\pi}{3}$ (ii) $z^{10} = (2 \operatorname{cis} \frac{2\pi}{3})^{10}$ $= 2^{10} (\cos \frac{20\pi}{3} + i \sin \frac{20\pi}{3})$ $= 2^{10} (\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3})$ ✓ $= 2^{10} (-\frac{1}{2} + \frac{\sqrt{3}}{2}i)$ $= 2^9 (-1 + \sqrt{3}i)$ ✓		(i)  $(x-3)^2 + (y-4)^2 = 2^2$ $x=0 \quad y-4 = \pm 4$ $y=0 \quad x=0, 6$ (ii) when P has max modulus it will be on diameter opposite origin $\arg(z) = \tan^{-1}(\frac{4}{3})$ ✓	✓ correct diagram ✓ correct intercepts
(f) $z - 3 - 4i$ $= z - (3 + 4i) = 5$ circle centre (3, 4) radius 5		12. (a) $\underline{r} = \underline{i} + 3\underline{j} - 9\underline{k} + t(\underline{i} + 12\underline{j} + 2\underline{k})$ $\underline{r} = \begin{pmatrix} 1 \\ 3 \\ -9 \end{pmatrix} + t \begin{pmatrix} 1 \\ 12 \\ 2 \end{pmatrix}$ Sphere $\left \begin{pmatrix} x \\ y \\ z \end{pmatrix} - \begin{pmatrix} 1 \\ 3 \\ -9 \end{pmatrix} \right = 9$ from $\underline{r}$ $x = 1 + t$ $y = 3 + 12t$ $z = -9 + 2t$	

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Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
Sub into eqn of sphere $(1+t-1)^2 + (3+2t-3)^2 + (-4+2t+4)^2 = 81$ $t^2 + 4t^2 + 4t^2 = 81$ $t^2 = 9$ $t = \pm 3$ ∴ Points of intersection are $t = 3 \begin{bmatrix} 4 \\ 9 \\ 2 \end{bmatrix}$ $t = -3 \begin{bmatrix} -2 \\ -3 \\ -10 \end{bmatrix}$		$x = 1$ $20 = 12 - 2\beta + 2$ $\beta = -3$ $I = \int \frac{4}{x-3} + \frac{-3x-1}{x^2+2}$ $= 4 \ln(x-3) - \frac{3}{2} \int \frac{2x}{x^2+2} - \int \frac{1}{x^2+2}$ $= 4 \ln(x-3) - \frac{3}{2} \ln(x^2+2) - \frac{1}{\sqrt{2}} \tan^{-1}\left(\frac{x}{\sqrt{2}}\right) + C$	
(b) $\frac{x^2+8x+11}{(x-3)(x^2+2)} = \frac{A}{x-3} + \frac{Bx+C}{x^2+2}$ $x^2+8x+11 = A(x^2+2) + (Bx+C)(x-3)$ $x=3$ $11A = 44 \quad A=4$ $x=0$ $11 = 2A - 3C$ $3C = 8 - 11$ $C = -1$		(c) $z = \frac{1}{1+is}$ $-\infty < s < \infty$ (i) $\frac{1}{1+is} = \frac{1}{2} + \frac{1}{2}i$ $\frac{2}{1+is} = 1+i$ $2 = (1+i)(1+is)$ $2 = (1-s) + i(1+s)$ $2 = 1-s \Rightarrow s = -1$ (ii) Let $z = x+iy$	

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Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
$\therefore x + iy = \frac{1}{1 + iS}$ $(x + iy)(1 + iS) = 1$ $x + yS + i(y + xS) = 1$ $x - yS = 1 \quad y + xS = 0$ $S = \frac{x-1}{y}$ $\therefore y + \frac{x(x-1)}{y} = 0$ $x^2 - x + y^2 = 0$ $(x - \frac{1}{2})^2 + y^2 = \frac{1}{4}$ $(x - \frac{1}{2})^2 + y^2 = (\frac{1}{2})^2$ circle centre $(\frac{1}{2}, 0)$ radius $r = \frac{1}{2}$	✓	$x = 2 + \lambda = -4 + 4u \quad \dots (1)$ $y = -3 + 4\lambda = 6 - 11u \quad \dots (2)$ $z = 1 + \lambda = -2 + 3u \quad \dots (3)$ from (3) $\lambda = -3 + 3u$ sub into (1) $2 + (-3 + 3u) = -4 + 4u$ $u = 3 \quad \& \quad \lambda = 6 \quad \checkmark$ check in (2) $LHS = -3 + 4(6) = 21 \quad \checkmark$ $RHS = 6 - 11(3) = -27$ $LHS \neq RHS$ $\therefore$ lines are skew	
(d) $\underline{x}_1 = \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 4 \\ 1 \end{pmatrix}$ $\underline{x}_2 = \begin{pmatrix} -4 \\ 6 \\ -2 \end{pmatrix} + \mu \begin{pmatrix} 4 \\ -11 \\ 3 \end{pmatrix}$		(e) $x \in \mathbb{R}$: if $x^2 - 6x + 5$ is even then x is odd. Using contrapositive let x be even ie $x = 2k \quad k \in \mathbb{Z}$ $\therefore x^2 - 6x + 5 = (2k)^2 - 6(2k) + 5$	

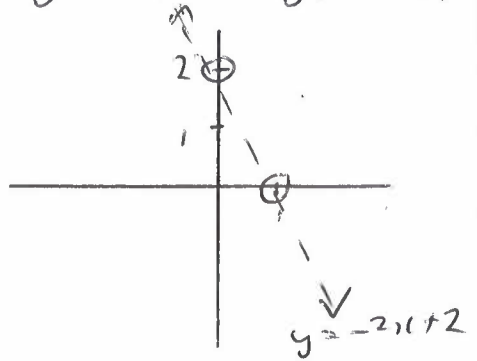
Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
$= 4k^2 - 12k + 4 + 1$ $= 2(2k^2 - 6k + 2) + 1$ $= 2p + 1 \quad p \in \mathbb{Z}$ $\therefore x^2 - 6x + 5$ is odd $\therefore$ by contrapositive if $x^2 - 6x + 5$ even then x is odd.		$\lambda = -2$ $\text{point} = \begin{pmatrix} 4 \\ 6 \end{pmatrix} + (-2) \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}$ $= \begin{pmatrix} -3 \\ 2 \\ 4 \end{pmatrix} \checkmark \text{ closest to origin}$	
13. (a) $\underline{x} = \begin{pmatrix} 4 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}$ point on $\underline{x}$ $x = 1 + 2\lambda$ $y = 4 + \lambda$ $z = 6 + \lambda$ distance from origin $d^2 = (1+2\lambda)^2 + (4+\lambda)^2 + (6+\lambda)^2$ $= 1 + 4\lambda + 4\lambda^2 + 16 + 8\lambda + \lambda^2 + 36 + 12\lambda + \lambda^2$ $= 6\lambda^2 + 24\lambda + 53$ $\frac{d d^2}{d\lambda} = 12\lambda + 24 = 0 \text{ at max/min}$ $\lambda = -2 \checkmark$		(b)  $v^2 = p + 8x - 9x^2$ $\dot{x} = \frac{d}{dt} (\frac{1}{2} v^2)$ $= \frac{d}{dt} (\frac{1}{2} p + 4x - \frac{9}{2} x^2)$ $= 4 - 9x$ $= -9 (x - \frac{4}{9})$ Centre at $x = 2$ $\therefore \frac{4}{9} = 2$ $9 = 2$ $h^2 = 9 = 2$ $h = \sqrt{2}$ period = $\frac{2\pi}{\sqrt{2}} = \sqrt{2}\pi$ (OR)	

$$\frac{d d^2}{d\lambda} = 12 > 0 \therefore \text{min}$$

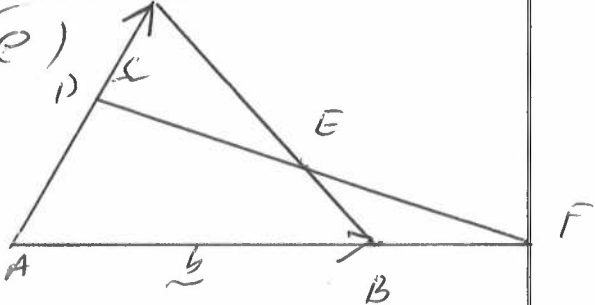
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Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
$v^2 = n^2(c^2 - (x-c)^2)$ $v^2 = n^2(25 - (11-2)^2)$ $= n^2(25 - 9^2 + 41 - 4)$ $= 2/n^2 + 4n^2 - n^2$ $p = 2/n^2 \quad q = n^2$ $4n^2 = 8$ $n = \pm \sqrt{2}$ $p = \sqrt{2}\pi$		(ii) $v \frac{dv}{dc} = -kv^3$ $\int_u^v \frac{1}{v^2} dv = \int_0^D -k dc$ $\left[-\frac{1}{v} \right]_u^v = \left[-kd \right]_0^D$ $-\frac{1}{v} + \frac{1}{u} = -kD$ $\frac{1}{v} - \frac{1}{u} = kD$	
(c) $\ddot{x} = -kv^3$ $t=0 \quad x=0 \quad v=u$ $t=T \quad x=D \quad v=V$ (i) $\frac{dv}{dt} = -kv^3$ $\int_u^v \frac{1}{v^3} dv = \int_0^T -k dt$ $\left[\frac{v^{-2}}{-2} \right]_u^v = \left[-kt \right]_0^T$ $-\frac{1}{2v^2} + \frac{1}{2u^2} = -kT$ $\frac{1}{v^2} - \frac{1}{u^2} = 2kT$		(iii) from (i) $\frac{1}{v^2} - \frac{1}{u^2} = 2kT$ $\left(\frac{1}{v} - \frac{1}{u} \right) \left(\frac{1}{v} + \frac{1}{u} \right) = 2kT$ $kD \left(\frac{v+u}{vu} \right) = 2kT$ $\frac{D}{T} = \frac{2vu}{u+v}$	
		(d) (i) $(a-b)^2 = a^2 + b^2 - 2ab \geq 0$ $a^2 + b^2 \geq 2ab$	

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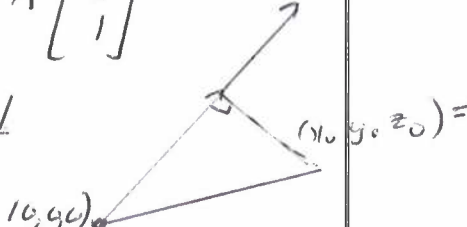
Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
Similarly $a^2 + c^2 \geq 2ac$ $b^2 + c^2 \geq 2bc$ $\therefore 2(a^2 + b^2 + c^2) \geq 2(ab + ac + bc)$ $a^2 + b^2 + c^2 \geq ab + ac + bc$ (ii) from (i) $x^2 + y^2 + z^2 \geq xy + yz + zx$ let $x = ab$ $y = ac$ $z = bc$ $(ab)^2 + (ac)^2 + (bc)^2 \geq a^2bc + ab^2c + abc^2$ $a^2b^2 + a^2c^2 + b^2c^2 \geq abc(a + b + c)$	✓ ✓	$n^3 + (n+1)^3 = (3k+1)^3 + (3k+2)^3$ $= 27k^3 + 27k^2 + 9k + 1 + 27k^3 + 54k^2 + 36k + 8$ $= 54k^3 + 81k^2 + 45k + 9$ $= 9[6k^3 + 9k^2 + 5k + 1]$ $= 9Q \quad Q \in \mathbb{Z}$ $\therefore n^3 + (n+1)^3$ is divisible by 9	3 3 ✓
Q14. (a) If $n, n+1$ are not divisible by 3 then let $n = 3k+1$ $n = 3k+2$ $k \in \mathbb{Z}$		(b) $w = \frac{z-1}{z-2i}$ if $w \in \mathbb{R} \Rightarrow$ $\arg(w) = 0 \text{ or } \pi$ $\arg(z-1) - \arg(z-2i) = 0, \pi$  $\therefore$ locus is the line $y = -2x + 2$ excluding $x=0$ and $x=1$	✓ ✓

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Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
(C) p is prime (i) a +ve integer $a \in \mathbb{Z}^+$ $(1+a)^p = 1 + \binom{p}{1}a + \binom{p}{2}a^2 + \dots + \binom{p}{p}a^p$ $\therefore (1+a)^p - 1 - a^p$ $= \binom{p}{1}a + \binom{p}{2}a^2 + \dots + \binom{p}{p-1}a^{p-1}$ $\binom{p}{k}$ is divisible by p $\therefore (1+a)^p - 1 - a^p$ is divisible by p (ii) $a^p - a$ let $a = 1$ $a^p - a = 1^p - 1 = 0 = p \times 0$ $\therefore$ true for $a = 1$ Assume true for $a = k$ $k^p - k = pM \quad M \in \mathbb{Z}$ let $a = k+1$ $\text{LHS} = (k+1)^p - (k+1)$ $= (k+1)^p - 1 - k$ $= (k+1)^p - 1 - k^p + k^p - k$ $= pQ + k^p - k \text{ from (i)} \quad Q \in \mathbb{Z}$ $= pQ + pM \text{ from assumption}$		$= pN ; N \in \mathbb{Z}$ $\therefore$ true for $a = k+1$ $\therefore$ true by Mathematical Induction ie $a^p - a$ is divisible by $p, a \in \mathbb{Z}^+$	
		(d) $\int \frac{x^2}{x+1} dx$ $= \int x - 1 + \frac{1}{x+1} dx$ $= \frac{x^2}{2} - x + \ln(x+1) + C$	
		(e)  AD:DC = 2:1 BE:EC = 1:2 (i) $\vec{CB} = \underline{b} - \underline{c}$ $\vec{CE} = \frac{2}{3} \vec{CB}$ $= \frac{2}{3} (\underline{b} - \underline{c})$	

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Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
(ii) $\vec{DE} = \vec{DC} + \vec{CE}$ $= \frac{1}{3}\vec{c} + \frac{2}{3}(\vec{b} - \vec{c})$ $= \frac{2}{3}\vec{b} - \frac{1}{3}\vec{c}$ (iii) $\vec{EF}$ is a scalar multiple of $\vec{DE}$ and $\vec{BF}$ is a scalar multiple of $\vec{AB}$ So $\vec{EF} = k \vec{DE} = k(\frac{2}{3}\vec{b} - \frac{1}{3}\vec{c})$ ✓ $\vec{BF} = l \vec{AB} = l\vec{b}$; $k, l \in \mathbb{Z}$ $\vec{EB} + \vec{BF} = \vec{EF}$ $\frac{1}{3}(\vec{b} - \vec{c}) + l\vec{b} = k(\frac{2}{3}\vec{b} - \frac{1}{3}\vec{c})$ ✓ $\frac{1}{3}\vec{b} - \frac{1}{3}\vec{c} + l\vec{b} = \frac{2k}{3}\vec{b} - \frac{k}{3}\vec{c}$ ✓ as $\vec{b}$ & $\vec{c}$ not parallel $\frac{1}{3} + l = \frac{2k}{3}$ $-\frac{1}{3} = -\frac{k}{3} \Rightarrow k=1$ ✓ $l = \frac{1}{3}$ $\vec{BF} = \frac{1}{3}\vec{AB}$ so $AF:FB = 4:1$		15(a) $\bar{L}_n = \int_0^1 (1-x^2)^{\frac{n-1}{2}} dx$ $\bar{L}_n = \left[x(1-x^2)^{\frac{n-1}{2}} \right]_0^1 - \frac{n-1}{2} \int_0^1 x(1-x^2)^{\frac{n-3}{2}} (-2x) dx$ $= 0 - (n-1) \int_0^1 (-x^2)(1-x^2)^{\frac{n-3}{2}} dx$ $= 0 - (n-1) \int_0^1 (1-x^2-1)(1-x^2)^{\frac{n-3}{2}} dx$ $= 0 - (n-1) \int_0^1 (1-x^2) + (n-1) \int_0^1 (1-x^2)^{\frac{n-3}{2}} dx$ $= 0 - (n-1)\bar{L}_n + (n-1)\bar{L}_{n-2}$ ✓ $n\bar{L}_n = x(1-x^2)^{\frac{n-1}{2}} + (n-1)\bar{L}_{n-2}$ ✓	
		(b) $F = \frac{m}{x^3} (6-10x)$ (i) $t=0$ $v=0$ $x=1$ $v \frac{dv}{dx} = \frac{1}{x^3} (6-10x)$ $\int_0^x v dv = \int_1^x \frac{6-10u}{u^3} du$ $\frac{v^2}{2} = \int_1^x (6u^{-3} - 10u^{-2}) du$ $= \left[-\frac{3}{u^2} + \frac{10}{u} \right]_1^x$ $= -\frac{3}{x^2} + \frac{10}{x} - 7$ ✓ $v^2 = -\frac{6}{x^2} + \frac{20}{x} - 14$ or $2\left(\frac{-3+10x-7x^2}{x^2}\right)$ ✓	

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$v = \frac{-1 \pm \sqrt{20x - 10x^2 - 6}}{x^2}$ $= -\frac{\sqrt{20x - 10x^2 - 6}}{x} \checkmark$ as $x=1$, $v=0$ and $\ddot{x} < 0$ $\therefore$ it moves in s-ve direction (ii) comes to rest when $v=0$ $\therefore 20x - 10x^2 - 6 = 0$ $7x^2 - 10x + 3 = 0$ $(7x-3)(x-1) = 0$ $x = 1 \text{ or } \frac{3}{7}$ ie at $x = \frac{3}{7}$ $\checkmark$		$= \left[\sqrt{-2+3x-x^2} \right]_1^2 + \frac{5}{2} \left[\sin^{-1}(2x-3) \right]_1^2$ $= \frac{5}{2} (\sin^{-1}1 - \sin^{-1}(-1))$ $= \frac{5\pi}{2} \checkmark$	
(d) $\int_1^2 \frac{x+1}{\sqrt{-2+3x-x^2}} dx$ $= -\frac{1}{2} \int_1^2 \frac{-2x+3-5}{\sqrt{-2+3x-x^2}} dx$ $= -\frac{1}{2} \int_1^2 \frac{-2x+3}{\sqrt{-2+3x-x^2}} dx + \frac{5}{2} \int_1^2 \frac{1}{\sqrt{(\frac{3}{2})^2 - (x-\frac{3}{2})^2}} dx$		(d) $\underline{r} = \lambda \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$ <u>Method 1</u>  let $\underline{s} = (x_0, y_0, z_0)$ $\text{proj}_{\underline{r}} \underline{s} = \frac{(x_0, y_0, z_0) \cdot (1, 2, 1)}{\sqrt{1+4+1} \sqrt{1+4+1}} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$ $= \frac{x_0 + 2y_0 + z_0}{6} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$ <u>Method 2</u> Point on $\underline{r} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$ distance to $\underline{s} = d$ $d^2 = (x_0 - \lambda)^2 + (y_0 - 2\lambda)^2 + (z_0 - \lambda)^2$ $= x_0^2 - 2x_0\lambda + \lambda^2 + y_0^2 - 4y_0\lambda + 4\lambda^2 + z_0^2 - 2z_0\lambda + \lambda^2$	

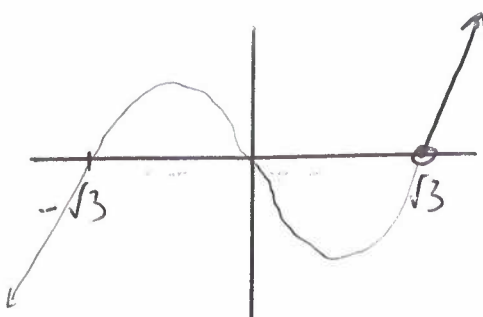
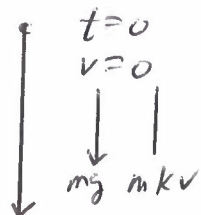
$$\frac{d}{d\lambda} d^2 = -2x_0 + 2\lambda - 4y_0 + 8\lambda - 2z_0 + 2\lambda = 0$$

$$12\lambda = 2x_0 + 4y_0 + 2z_0$$

$$\lambda = \frac{x_0 + 2y_0 + z_0}{6}$$

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
point on Σ closest to (x_0, y_0, z_0) $= \frac{x_0 + 2y_0 + z_0}{6} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$ (ii) $\Sigma = \begin{bmatrix} 4 \\ 1 \\ 6 \end{bmatrix} + t \begin{bmatrix} -1 \\ 1 \\ 5 \end{bmatrix}$ $= \begin{bmatrix} 4-t \\ 1+t \\ 5t \end{bmatrix}$ from (i) above closest point on Σ will be $\frac{4-t + 2(1+t) + 5t}{6} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$ $= (1+t) \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \checkmark$ So point on Σ that has a perpendicular		distance to a point on Σ is $(t+1, 2t+2, t+1)$ Point on Σ is $(4-t, 1+t, 5t)$ distance between points ℓ $\ell^2 = (4-t-t-1)^2 + (1+t-2t-2)^2 + (5t-t-1)^2$ $= 21t^2 - 18t + 11 \checkmark$ $\frac{d\ell^2}{dt} = 42t - 18 = 0 \text{ at min}$ $t = \frac{3}{7}$ $\frac{d^2\ell^2}{dt^2} = 42 > 0 \therefore \text{min} \checkmark$ position vector is $\left(1 + \frac{3}{7}\right) \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$ $= \begin{bmatrix} \frac{10}{7} \\ \frac{20}{7} \\ \frac{10}{7} \end{bmatrix} \text{ or } \frac{10}{7} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$	

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Q16 (9) (i) $\int_n^{2n} \frac{dx}{\sqrt{x}}$ $= \left[2\sqrt{x} \right]_n^{2n}$ $= 2\sqrt{2n} - 2\sqrt{n}$ $= 2\sqrt{n}(\sqrt{2}-1)$		$S_n < 2\sqrt{n}(\sqrt{2}-1) < \frac{1}{\sqrt{n}} + \left(\frac{1}{\sqrt{n+1}} + \frac{1}{\sqrt{n+2}} + \dots + \frac{1}{\sqrt{2n-1}} + \frac{1}{\sqrt{2n}} \right) - \frac{1}{\sqrt{2n}}$ $S_n < 2\sqrt{n}(\sqrt{2}-1) < \frac{1}{\sqrt{n}} + S_n - \frac{1}{\sqrt{2n}}$ $0 < 2\sqrt{n}(\sqrt{2}-1) - S_n < \frac{1}{\sqrt{n}} - \frac{1}{\sqrt{2n}}$ $-2\sqrt{n}(\sqrt{2}-1) < -S_n < \frac{1}{\sqrt{n}} - \frac{1}{\sqrt{2n}} - 2\sqrt{n}(\sqrt{2}-1)$ $2\sqrt{n}(\sqrt{2}-1) > S_n > -\frac{1}{\sqrt{n}} + \frac{1}{\sqrt{2n}} + 2\sqrt{n}(\sqrt{2}-1)$ ie $2\sqrt{n}(\sqrt{2}-1) + \frac{1-\sqrt{2}}{\sqrt{2n}} < S_n < 2\sqrt{n}(\sqrt{2}-1)$	
(ii) Area of lower rectangle < Area under curve < Area of upper rectangles $\frac{1}{\sqrt{n+1}} + \frac{1}{\sqrt{n+2}} + \dots + \frac{1}{\sqrt{n}} < \int_n^{2n} \frac{dx}{\sqrt{x}}$ $\frac{1}{\sqrt{n}} + \frac{1}{\sqrt{n+1}} + \dots + \frac{1}{\sqrt{2n-1}}$		(b) $r = \begin{pmatrix} t + \frac{1}{t} \\ t^3 + \frac{1}{t^3} \end{pmatrix} \quad t \in \mathbb{R}^+$ $x = t + \frac{1}{t}$ $y = t^3 + \frac{1}{t^3}$ now $(t + \frac{1}{t})^3 = t^3 + 3t^2 \cdot \frac{1}{t} + 3t \cdot \frac{1}{t^2} + \frac{1}{t^3}$ $= (t^3 + \frac{1}{t^3}) + 3(t + \frac{1}{t})$ $= y + 3x$ $\therefore x^3 = y + 3x$ $y = x^3 - 3x = x(x-\sqrt{3})(x+\sqrt{3})$	

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 NOTE: $t \in \mathbb{R}^+$ $\therefore x > 0$ and $y > 0$ $\therefore y = x^3 - 3x = x(x - \sqrt{3})(x + \sqrt{3})$ with $x > 0, y > 0$.		(ii) $\frac{dv}{dt} = g - kv$ $= k(u - v)$ $\frac{1}{u - v} dv = k dt$ $\int_0^v \frac{1}{u - v} dv = \int_0^t k dt$ $\left[-\ln(u - v) \right]_0^v = \left[kt \right]_0^t$ $-\ln(u - v) + \ln u = kt$ $\ln\left(\frac{u}{u - v}\right) = kt$ $-\ln\left(\frac{u - v}{u}\right) = kt$ $\frac{u - v}{u} = e^{-kt}$ $u - v = ue^{-kt}$ $v = u(1 - e^{-kt})$ (iii) For particle A $\frac{dx}{dt} = u(1 - e^{-kt}) \text{ from (i)}$ $x_A = \int_0^T u(1 - e^{-kt}) dt$ $t = T \quad x = X$ $X = \int_0^T u(1 - e^{-kt}) dt$	
(c) (i)  $F_{\text{net down}} = mg - mkv$ $m\ddot{x} = mg - mkv$ $\ddot{x} = g - kv$ terminal velocity $\Rightarrow$ $\ddot{x} = 0$ $\therefore \frac{v}{T} = \frac{g}{k} = u$ $\ddot{x} = k\left(\frac{g}{k} - v\right)$ $\ddot{x} = k(u - v)$			

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For particle B $v = ke^{-kt} - u(1 - e^{-kt})$ $\frac{dx}{dt} = ke^{-kt} - u(1 - e^{-kt})$ $x_B = \int (ke^{-kt} - u(1 - e^{-kt})) dt$ $t = T \quad x_B = h - x$ $h - x = \int_0^T ke^{-kt} dt - \int_0^T u(1 - e^{-kt}) dt$ $h - x = u \int_0^T ke^{-kt} dt - x \quad \checkmark$ $h = \left[-ue^{-kt} \right]_0^T$ $h = -ue^{-kT} + u \quad \checkmark$ $ue^{-kT} = u - h$ $e^{-kT} = \frac{u - h}{u}$ $T = -\frac{1}{k} \ln \left(\frac{u - h}{u} \right)$ $= \frac{1}{k} \ln \left(\frac{u}{u - h} \right)$			